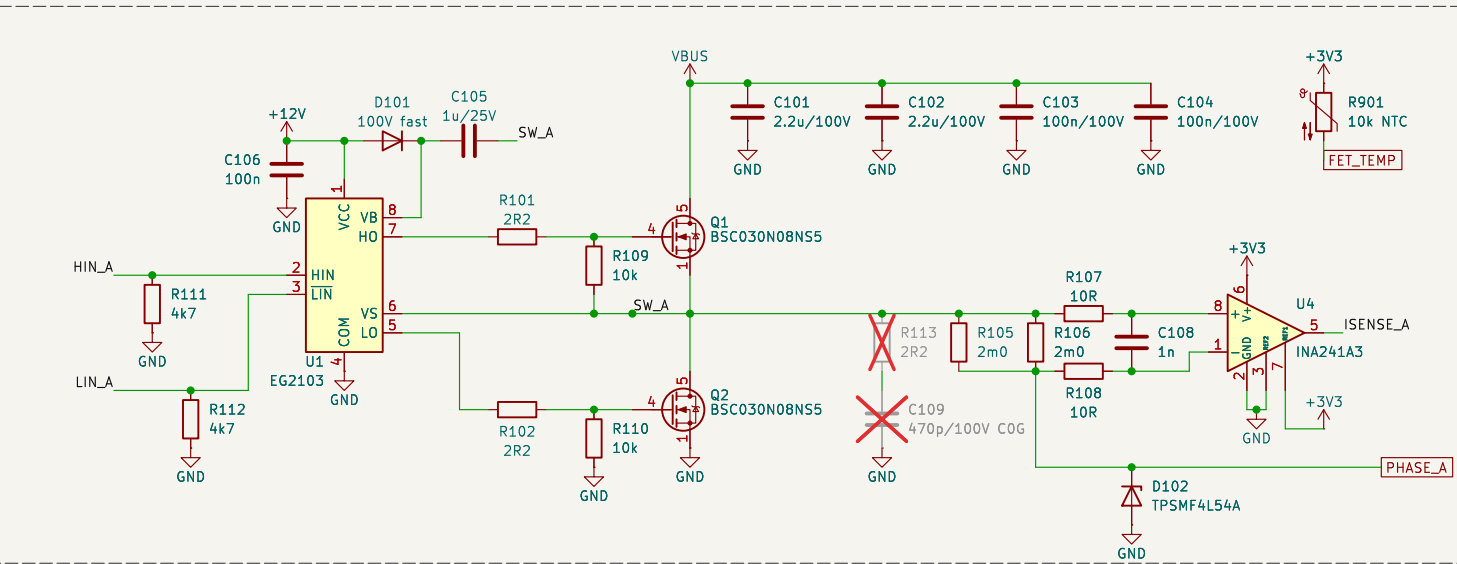
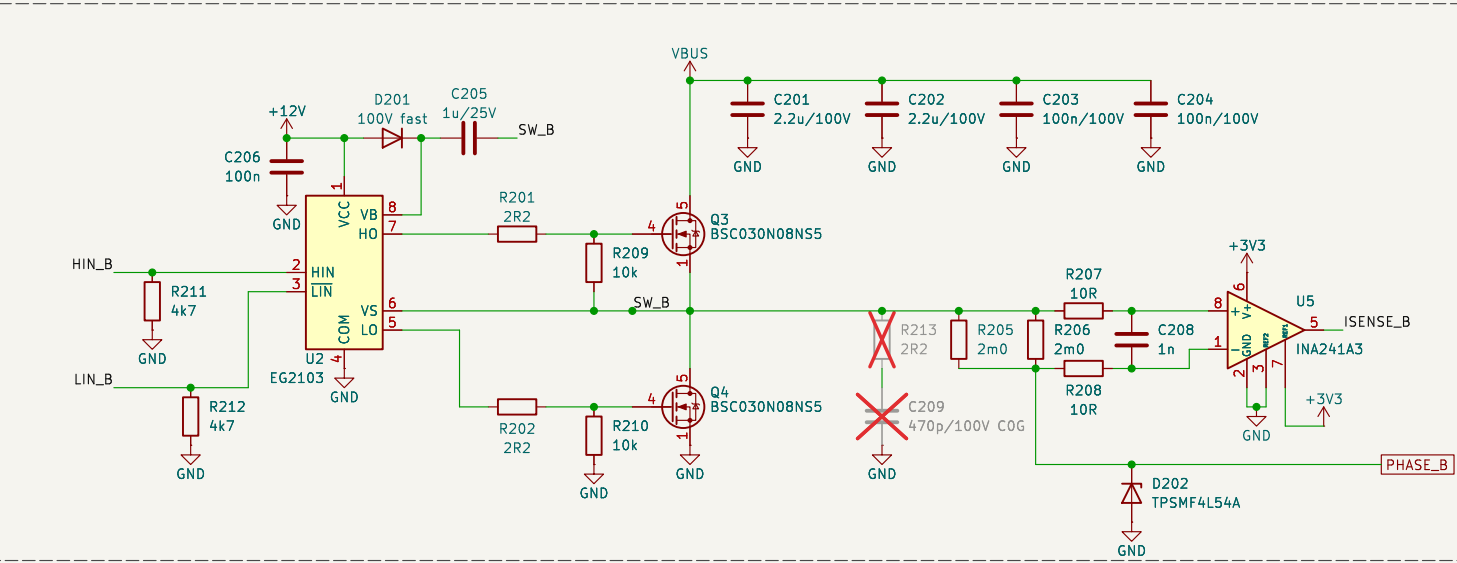




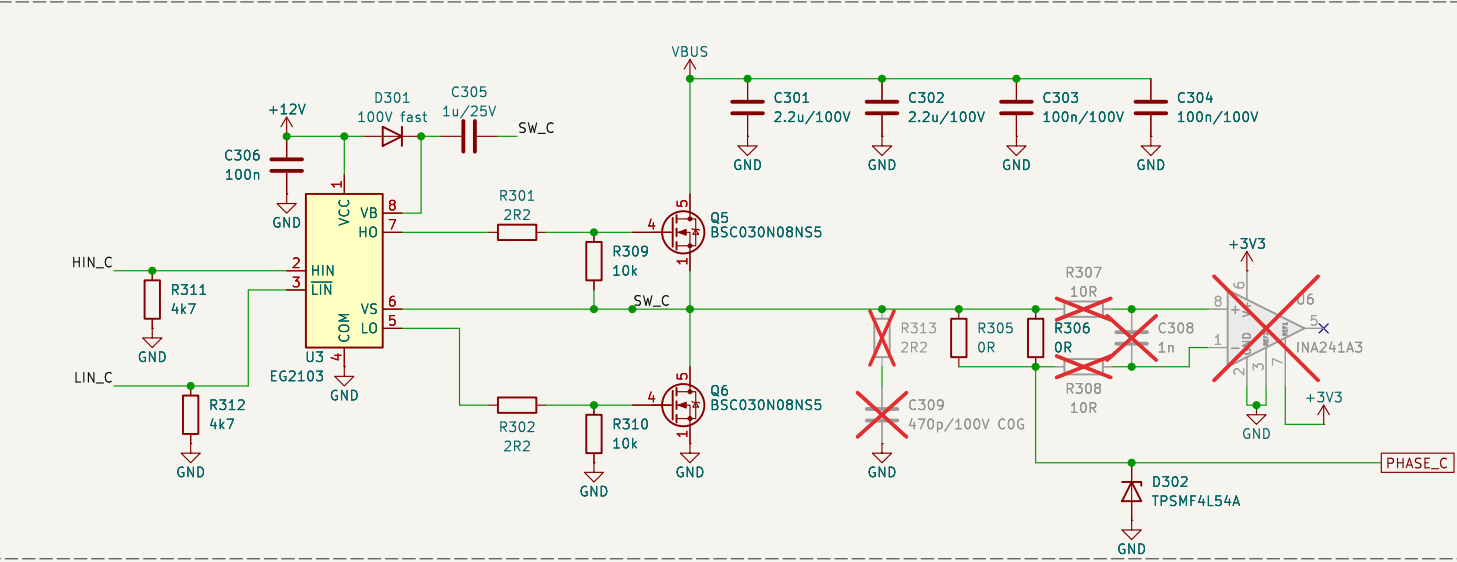
Phase A



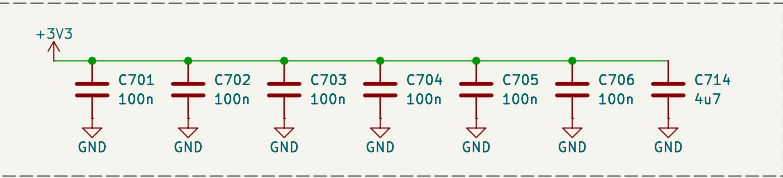
Phase B



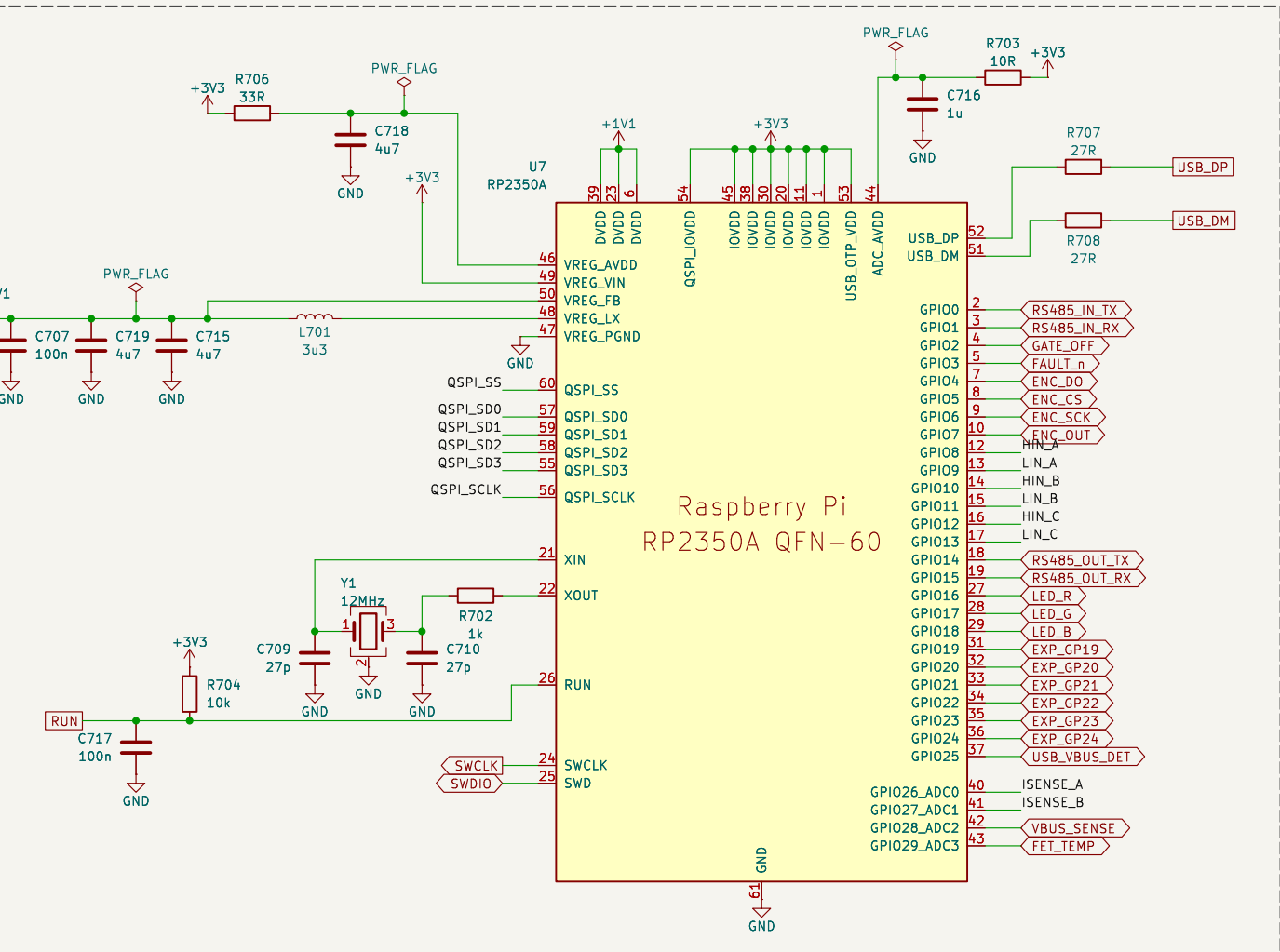
Phase C



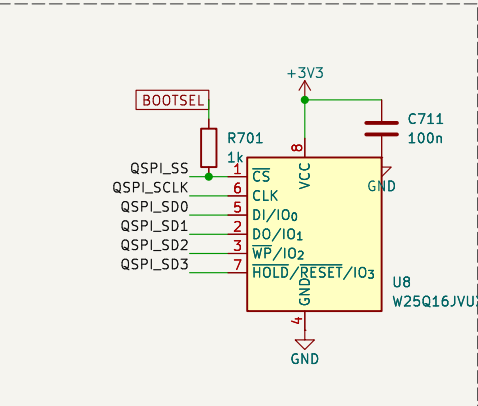
Decoupling



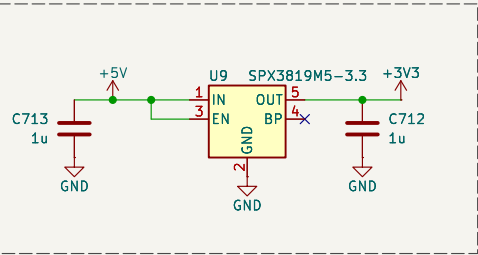
RP2350A



QSPI flash



3V3 LDO



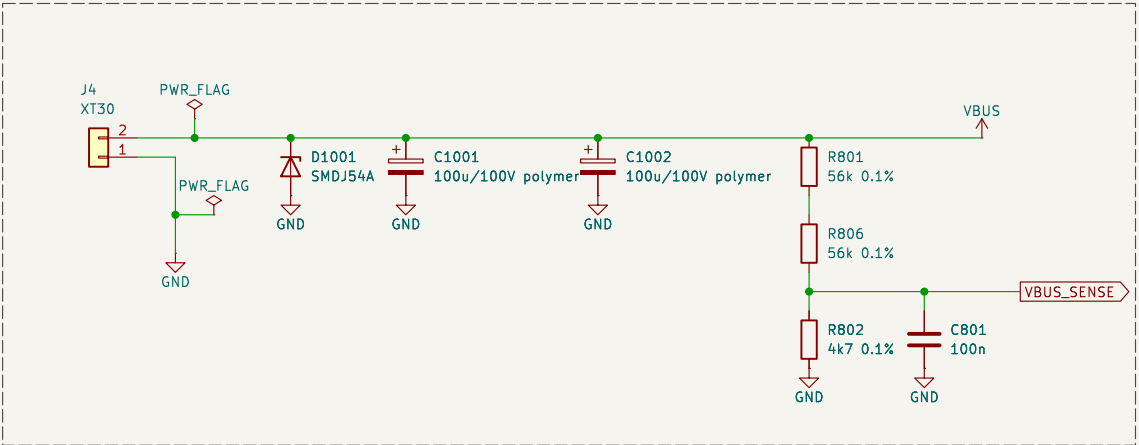
Notes: the power bridge

Each phase is an EG2103 half-bridge driver on a pair of BSC030N08N55: bootstrap diode and capacitor (D101, C105 on phase A), 2R2 gate resistors, 10k gate-source pull-downs.
HIN and LIN are held low (R111, R112): with both low the EG2103 turns the low side on, so a CPU in reset brakes the motor.
Inline shunts two 2mΩ in parallel between the switch node and the motor lead, Kelvin-sensed through 10R, 10R and 1n into an INA241A3. REF1 on +3V3 and REF2 on GND put its output at mid-rail, so it reads current either way.
Phase C: the shunts are 0R links and U6, R307, R308, C308 are not fitted. The RP2350A has no fourth ADC channel; phase C current is $-(I_a + I_b)$.
R113 + C109 (and B, C): RC snubber across each low side, footprints only.
R901: NTC beside the FETs, read on ADC3 as FET_TEMP.

Notes: the CPU

RP2350A as Raspberry Pi's minimal design (RP-006440): the core supply is the chip's own switching regulator, VREG_LX through L701 to +1V1, which VREG_FB senses and DVDD takes. VREG_AVDD is +3V3 through R706 and C718.
One 100n at each IOVDD pin (C701–C706): C714 4u7 at VREG_VIN. ADC_AVDD is +3V3 filtered by R703 and C716 (+3V3A).
BOOTSEL: SW1 on sheet 03 pulls QSPL_SS low through R701 at reset.
The encoder: ENC_DO on GPIO4, ENC_SCK on GPIO6, ENC_CS on GPIO5, ENC_OUT on GPIO7 — SPI0's RX and SCK, so SPI0 reads it.
GPIO19–24 go to the expansion header (sheet 03).

Bus input



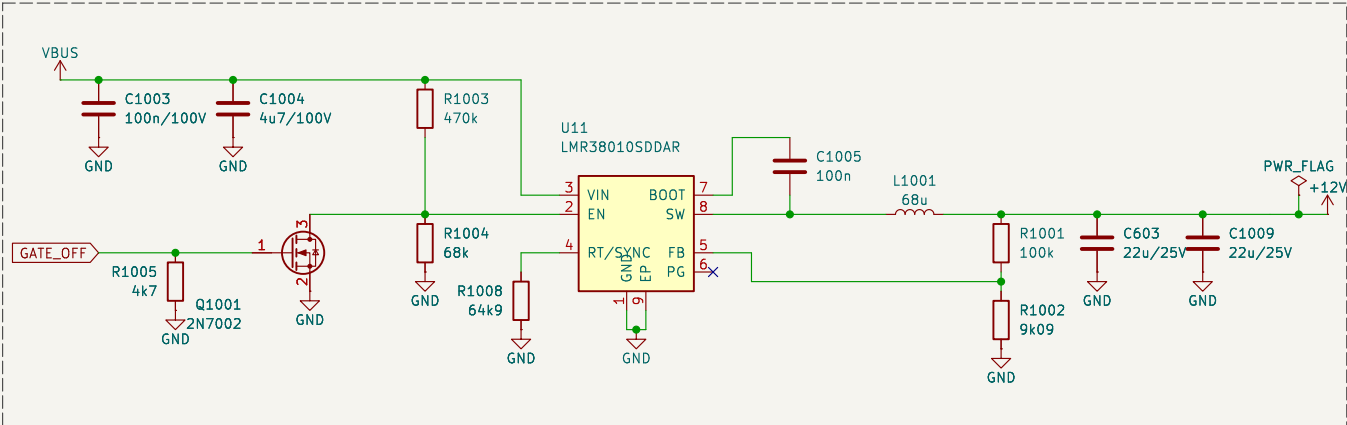
Notes

Bus: 48 V operating on 80 V FETs. D1001 (SMDJ54A, 54 V standoff) clamps it; C1001 and C1002 are the bulk.
 $VBUS_SENSE = VBUS \times 4k7 / 116k7$ (0.1 %), on ADC2.

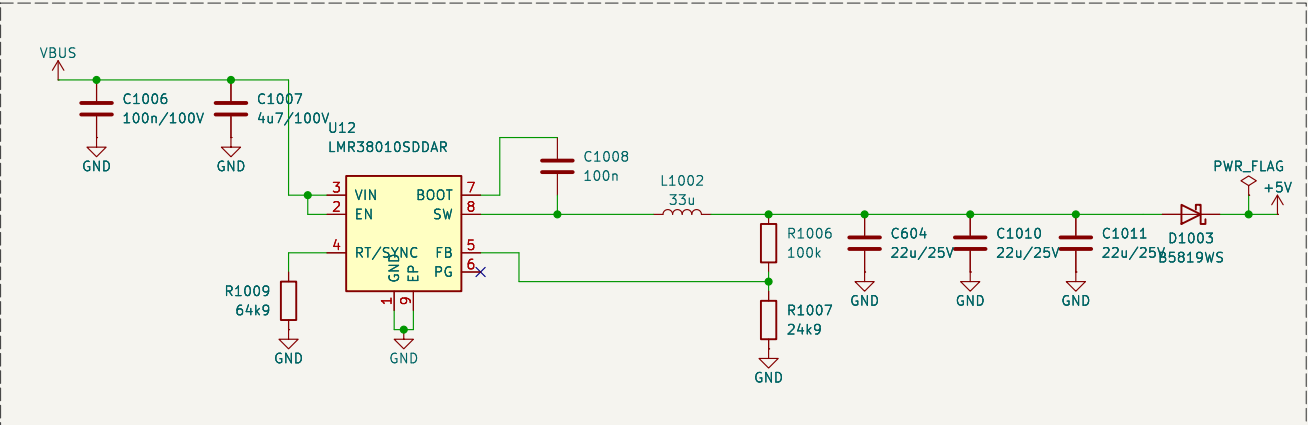
12 V gate rail: LMR38010, starting at 9.9 V on the R1003/R1004 divider. GATE_OFF (GPIO14) high turns Q1001 on and stops the rail, taking the gate drivers' supply away. R1005 holds it low through reset, so the rail comes up and the drivers brake the motor.

5 V logic rail: LMR38010, diode-ORed with USB (D1002, sheet 03) through D1003. The 3V3 LDO on sheet 01 runs from +5V.

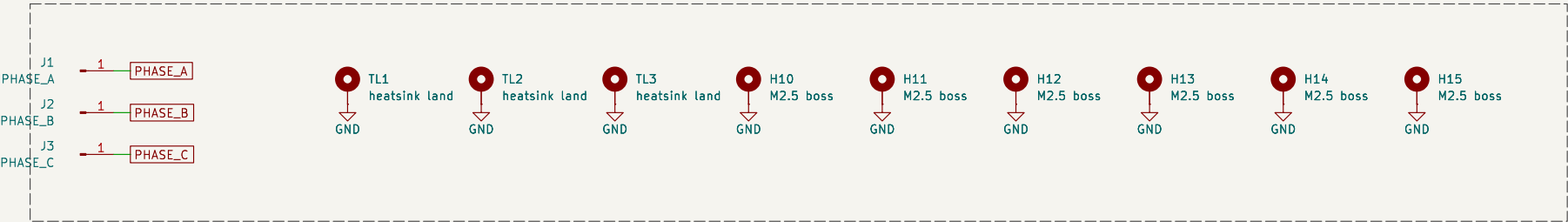
12 V gate rail



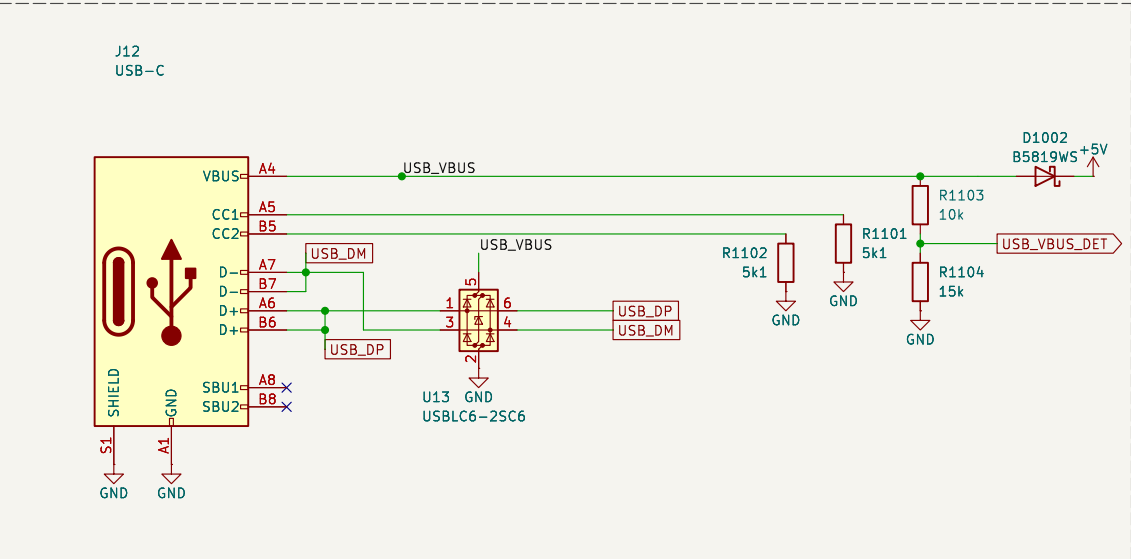
5 V logic rail



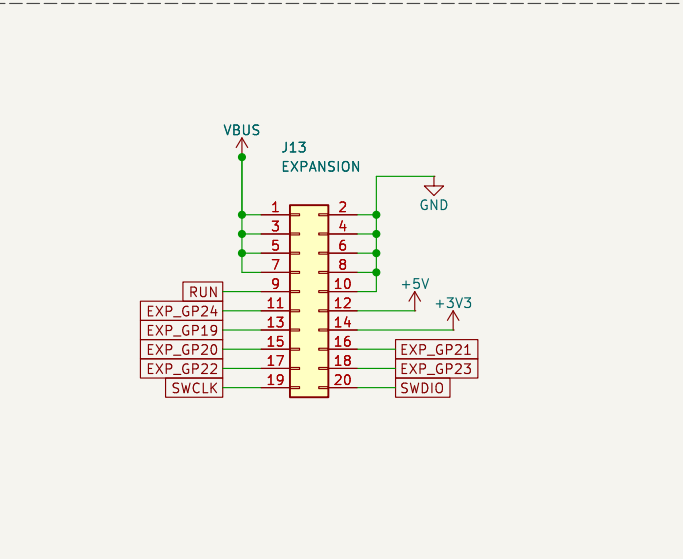
Mounting and motor leads



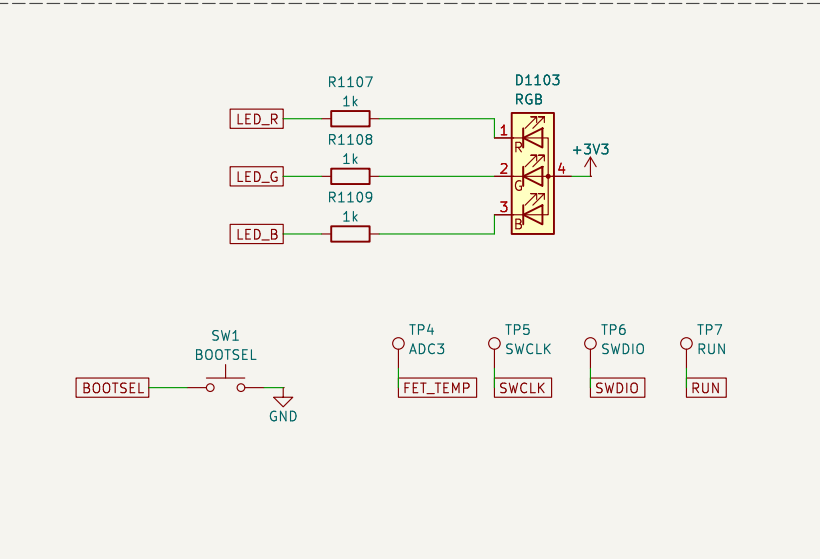
USB-C: data and 5 V logic, no PD



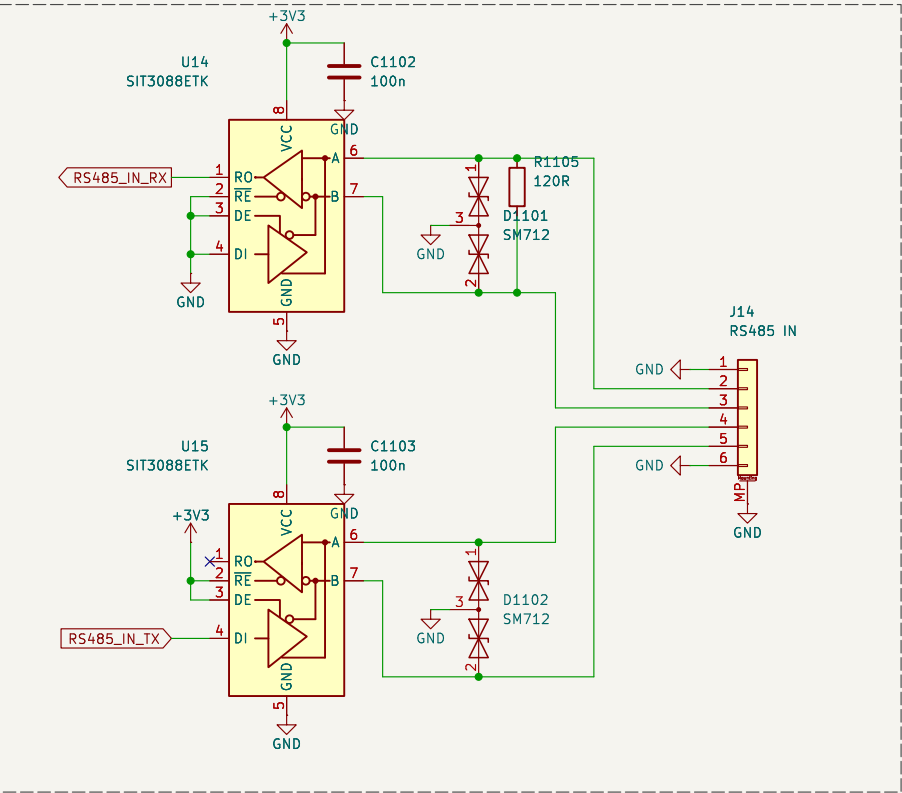
Expansion header: PD or Ethernet board



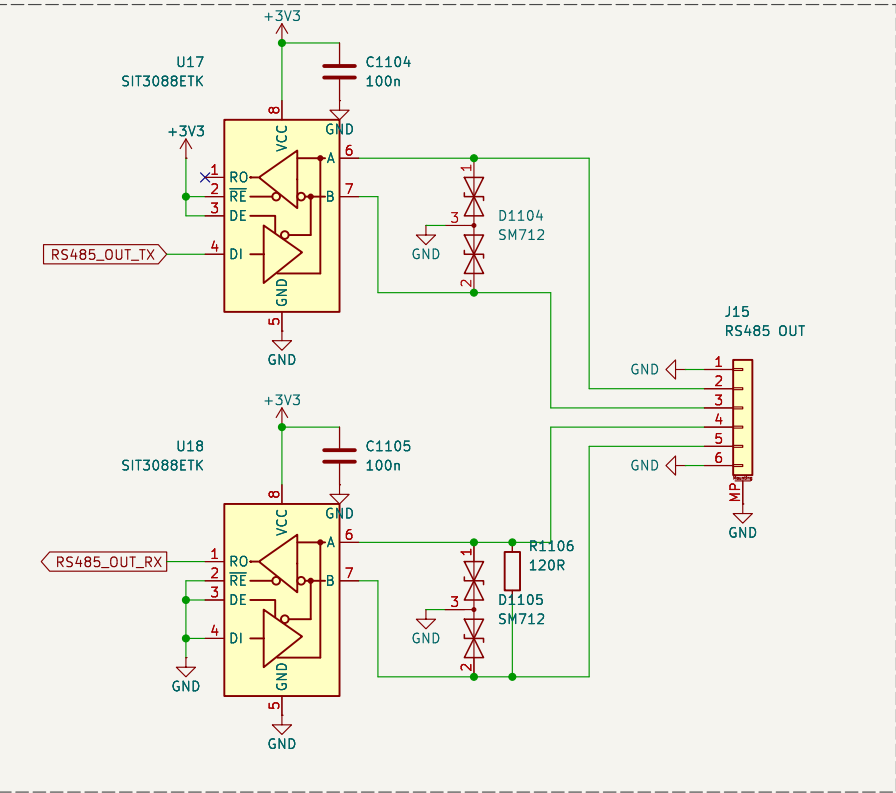
Status LED, BOOTSEL, test pads



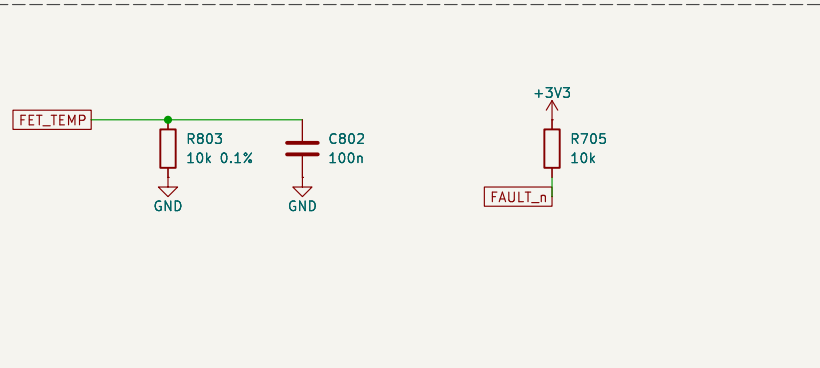
RS-485 relay: port IN



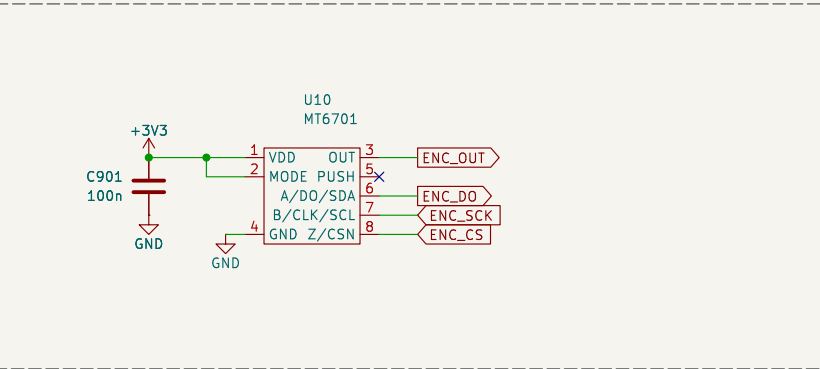
RS-485 relay: port OUT



ADC3 divider, FAULT_n pull-up



Encoder: MT6701 on the shaft axis



RS-485 is a CPU relay: each port is two point-to-point pairs, one in and one out, so a drive passes the chain on in software. Drivers are always enabled; each receiver has its 120R termination (R1105, R1106).

The MT6701 sits on the motor-facing side, centred on the shaft over a diametric magnet, MODE tied to VDD; the CPU reads it over SSI (sheet 01).

R803 with the NTC (R901, sheet 01) makes FET_TEMP.

CERN-OHL-P
 Drawn by tools/schlayout_s.py from the netlist in tools/schematic_s.py
Sequoia Hope Alexander
 Sheet: /03_encoder_io/
 File: 03_encoder_io.kicad_sch

Title: servodrive – board S, single board – 03_encoder_io

Size: A3 Date: 2026-09-25 Rev: A
 KiCad E.D.A. 9.0.8 Id: 4/4